LECTURER: JOHN DOE

REINFORCEMENT LEARNING

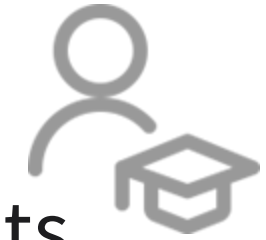
REINFORCEMENT LEARNING
TOPIC OUTLINE

Introduction to Reinforcement Learning	1
Dynamic Programming and Markov Decision Processes	2
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UNIT 5

REINFORCEMENT LEARNING APPROACHES

STUDY GOALS



- Understand **model-free reinforcement learning** concepts and terminology

- Explain the principles behind **model-based reinforcement learning** and its utility

- Assess and differentiate **exploration-exploitation strategies**, evaluating their strengths and weaknesses

EXPLAIN SIMPLY



1. In what situations does model-free reinforcement learning excel, and what makes it a valuable approach?

2. How do predictive models improve decision-making in uncertain environments within model-based reinforcement learning?

3. What strategies and trade-offs impact learning in addressing the exploration-exploitation dilemma?

Goal



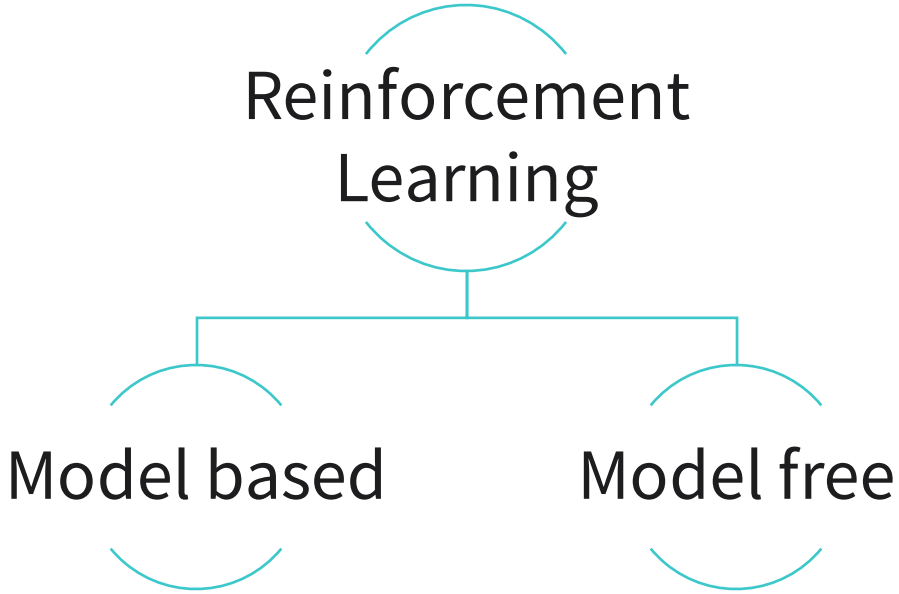
- Optimize decision-making process in reinforcement learning

Challenges



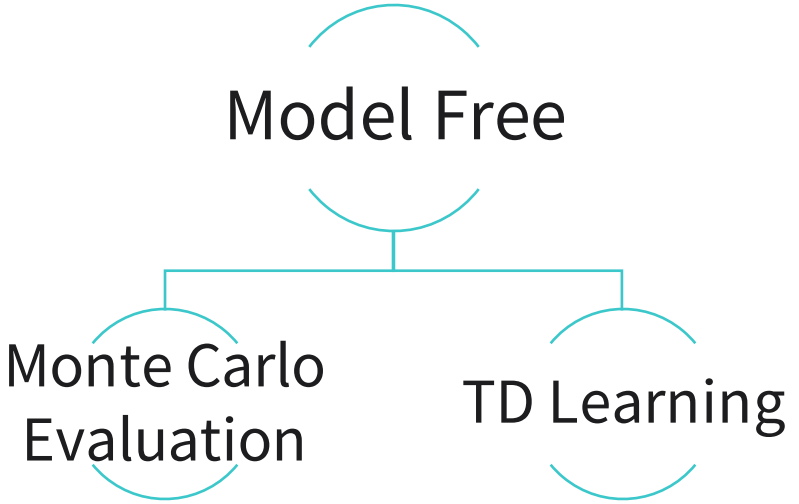
- Uncertainty in transition probabilities and rewards
- Scaling issues with large state/action spaces
- Exploration and learning from action

Fundamental Approaches



Model-free learning adapts in unknown, stochastic environments without explicit transition and reward models

- **Model-free prediction** estimates the value function
- **Model-free control** optimizes the value function
- Learn optimal actions directly from **experiences** in unknown environments



Experience tuple: (s, a, r, s')

MONTE CARLO POLICY EVALUATION

Evaluate policies directly from experience

No need to know the structure of the underlying problem



Model Free

1. **Initialize**
   - Policy π to evaluate
   - Arbitrary state-value function V
   - Returns for all states $Returns(s)$
2. **Repeat**
   - Use policy π to generate episode $(s_0, a_0, r_1, \ldots, s_{k-1}, a_{k-1}, r_k)$
   - For each state s in episode
     - Retrieve return r_i for s and append to list of returns
     - Compute value function

$$V(s) = \frac{1}{N} \sum_j Returns(s)_j$$

Source of the text: Sutton & Barto, 2018.

TEMPORAL DIFFERENCE LEARNING

- Learn without prior knowledge, interact with environment, no transition data
- TD learning is online and does not require complete episodes
- Prominent approaches: Q-learning and SARSA

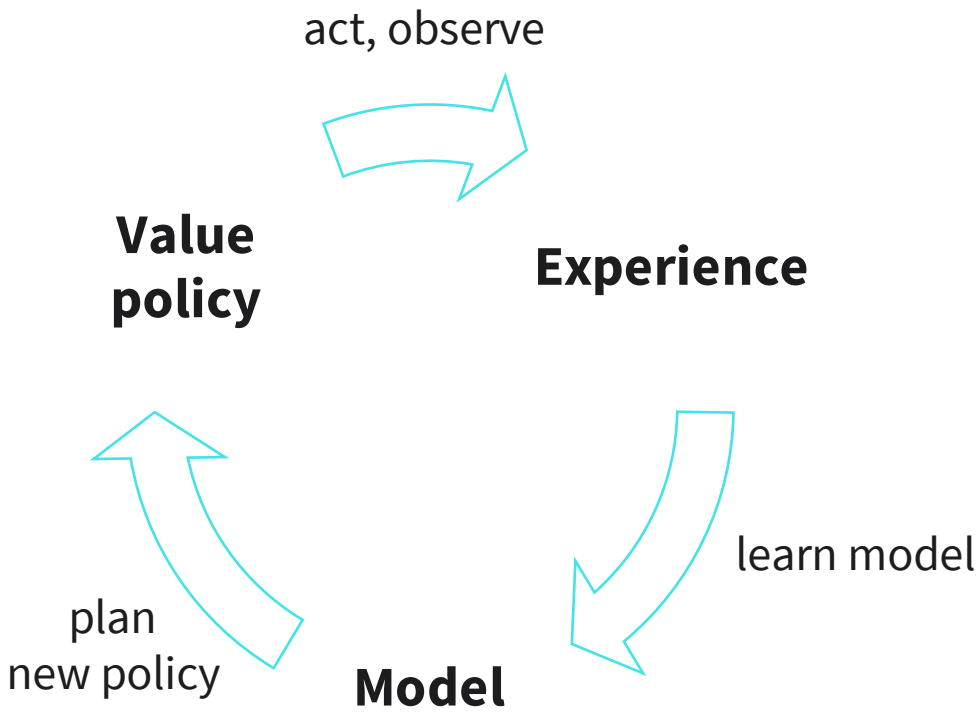
1. Initialize Q-values arbitrarily for all state-action pairs
2. Choose an action that maximizes $Q(s, a)$ in the current state
3. Observe the new state and the actual reward obtained
4. Update Q-values iteratively

$$Q(s,a) + \alpha \left[r + \gamma \max_{\alpha} Q(s', \alpha) - Q(s, \alpha) \right]$$

Source of the text: Watkins, 1989; Sutton & Barto, 2018.

Interact with the environment to gain experience and use it to build a model to devise optimal policies

- Model: state space S, action-space A, transition probabilities $P(s' \mid s, a)$ and rewards R
- Not known, depend on parameter θ, i.e., current environment knowledge
- Model parameters are learned from experience tuples $(s_0, a_0, r_1, \ldots, s_{k-1}, a_{k-1}, r_k)$



How do we learn model parameters from experience episodes?



Look Up Table

- Store experience tuples
- Use table for state-action pair lookup
- Not efficient for complex problems

Ready Policy One

- Build optimal world model
- Choose action, observe, update model with reward
- Use ready policy one to gather data
- Discrepancies with world model improve policy

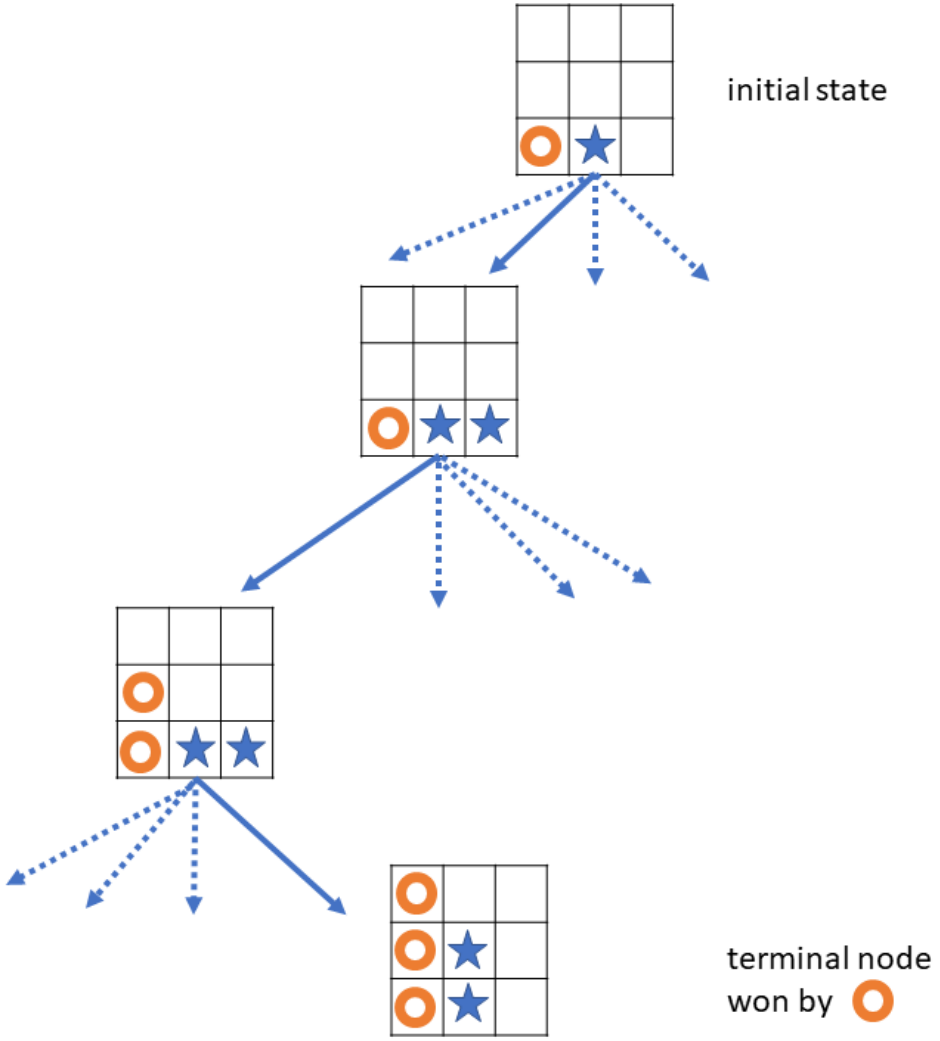
Source of the text: Sutton, 1990; Ball et.al., 2020.

Issues in long-term reward prediction

- Waiting for rewards in lengthy trajectories
- Exploring all possible moves is infeasible

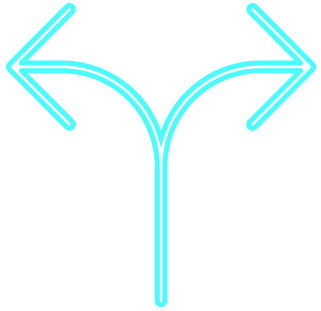
Approach

- **Select** tree for traversal, explore nodes
- **Expand** tree with child nodes
- **Simulate** environment for child nodes
- **Backpropagate** simulation results to update metrics



Source of the text: Coulom, 2006.
Source of the image: Own representation

EXPLORATION VERSUS EXPLOITATION



Exploitation

Should we stick with our known but maybe mediocre results?

- **Greedy:** choose action that gives highest result with current knowledge
- Exploit the environment

Exploration

Or risk something new with unknown rewards?

- ϵ **– Greedy:** introduce small probability for random actions
- Explore the environment

$$\alpha_t = \begin{cases} \alpha_t^* : & with\ probability\ 1 - \epsilon \\ \alpha_t^* : random\ action\ with\ probability\ \epsilon \end{cases}$$

Source of the text: Sutton & Barto, 2018.

EXPLORATION VERSUS EXPLOITATION APPROACHES FOR MODEL FREE LEARNING

Boltzmann equation

- Action choice based on Boltzmann probability
- Temperature parameter controls exploration vs. exploitation

$$P(a) = \frac{e^{\frac{Q(s,a)}{T}}}{\sum_{\alpha \in A} e^{\frac{Q(s,a)}{T}}}$$

Thompson sampling

- Bayesian approach; actions based on posterior distribution
- Modify distribution with observations, guide further actions
- Challenge (in complex scenarios): posterior (i.e., success rate) needs to be calculated

Source of the text: Thompson, 1933; Cesa-Bianchi et al, 2017.

EXPLORATION VERSUS EXPLOITATION APPROACHES FOR MODEL BASED LEARNING

Explicit **E**xplore or **E**xploit

- Distinguishes between exploration and exploitation phases
- Separates environment into known and unknown states
- Optimal policy in known states, learn in unknown

R - Max

- Separates MDP into known and unknown states
- Uses a fixed reward until states are known

$$R(s,a) = \begin{cases} \hat{R}(s,a) & \text{for known } (s,a) \\ \hat{R}_{max} & \text{for unknown } (s,a) \end{cases}$$

Source of the text: Brafman & Tennenholtz, 2002; Kearns & Singh, 2002.

REVIEW STUDY GOALS



- Understand **model-free reinforcement learning** concepts and terminology

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SESSION 5

TRANSFER TASK





Case study

Imagine you are a data scientist tasked with optimizing the delivery process for a food delivery service. The goal is to strike the right balance between exploring new delivery routes and exploiting known, efficient routes to ensure timely and cost-effective deliveries

Task

Develop an exploration-exploitation plan for food delivery efficiency balancing route optimization, data, and real-time decisions. What key steps will you consider?

TRANSFER TASK
PRESENTATION OF THE RESULTS

Please present your results.

The results will be discussed in plenary.





1. A reinforcement learning application has a state space $\chi = \{1,2,3,\ldots,100\}$. The action space is $A = \{1,2,\ldots,10\}$. State $x = 25$ was input into the Environment for a total of 132 stages. For 75% of those stages, action $a = 3$ was also input. Of all the stages in which state-action pair (25,3) was input into the Environment, 9 resulted in a transition to state 43. Give an estimate of $P(34 \mid 25, 3)$, where $P(y \mid x,a)$ is the transition probability kernel. Select the closest answer?
   a) 0.051
   b) 0.068
   c) 0.091
   d) 0.75



2. SARSA can be described as a …
   a) TD(0) on-policy control algorithm
   b) TD(0) off-policy prediction algorithm
   c) TD(1) on-policy prediction algorithm
   d) TD(1) on-policy control algorithm



3. An on-policy reinforcement learning control algorithm will use an ϵ-greedy policy. On stage k, the policy will choose an action at random with probability ϵ[k], otherwise it will use the current estimate of the optimal policy. The objective is to allow the greedy policy to become increasingly dominant while ensuring there is sufficient exploratory behavior to identify the optimal policy. Which of the following sequences for ϵ[k] will achieve this?

   a) ϵ[1], ϵ[2], ϵ[3] , … = 0.01, 0.01, 0.01, …
   b) ϵ[1], ϵ[2], ϵ[3] , … = (1/2)**1, (1/2)**2, (1/2)**3, …
   c) ϵ[1], ϵ[2], ϵ[3] , … = 1/1, 1.2, 1/3, …
   d) ϵ[1], ϵ[2], ϵ[3] , … = (0.99)**1, (0.99)**2, (0.99)**3, …

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